

CARBOXYLATED NBR (XNBR) SYNTHESIS AND POST-FUNCTIONALIZATION PATENT REVIEW

1. ABSTRACT

This report reviews patent evidence relevant to carboxylated nitrile rubber (XNBR), with emphasis on how carboxyl functionality is introduced into an NBR backbone and how the resulting material is subsequently processed. The single primary patent in the manifest, US20030109642A1, does not describe the initial emulsion copolymerization of XNBR; instead, it addresses a downstream transformation of carboxylated nitrile rubber, including hydrogenated carboxylated nitrile rubber (HXNBR), by simultaneous crosslinking and derivatization in the presence of coupling/condensing agents. The disclosed chemistry is centered on reacting the carboxyl-containing polymer with carbodiimides and related condensing agents to form crosslinked/derivatized networks while preserving the polymer backbone. The patent explicitly discusses polymers derived from conjugated dienes, unsaturated nitriles, and unsaturated carboxylic acids, and it highlights the challenge of maintaining carboxyl functionality during hydrogenation and subsequent processing.

The selected patent landscape therefore captures a key post-polymerization route for XNBR-derived materials rather than a direct synthesis route. The major process trend is the use of coupling/condensing agents, especially dicyclohexylcarbodiimide, to convert carboxyl groups into derived functionalities while simultaneously crosslinking the rubber. The evidence also shows that the polymer may be optionally hydrogenated, with selective hydrogenation of carbon-carbon double bonds emphasized to avoid reduction of nitrile and carboxyl groups. Formulation/process conditions disclosed in the examples include milling at 40°C, incorporation of 0.9 g or 2.4 g DCC into 24 g or 20 g HXNBR, and pressing at 150°C and 40,000 psi for 2 minutes in one example. Overall, the patent is technically relevant because it demonstrates a practical route for stabilizing and modifying XNBR/HXNBR after carboxyl incorporation, which is important for performance tuning in seals, hoses, belts, and other demanding rubber applications.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US20030109642A1
- Patent Title: Process for crosslinking carboxylated nitrile rubber, hydrogenating ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2003
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses simultaneous crosslinking and derivatization of a polymer containing units from a conjugated diene, an unsaturated nitrile, and an unsaturated carboxylic acid in the presence of a coupling/condensing agent. The process is applicable to optionally hydrogenated polymers, including hydrogenated carboxylated nitrile rubber (HXNBR), and uses carbodiimides such as dicyclohexylcarbodiimide to react with carboxyl functionality while forming a crosslinked material. - Polymer mass: 24 g — Example 1 - Dicyclohexylcarbodiimide (DCC) loading: 0.9 g — Example 1 - Milling temperature: 40° C. — Example 1 - Pressing temperature: 150° C. — Example 1 - Pressing pressure: 40,000 psi — Example 1 - Pressing time: 2 minutes — Example 1 - Polymer mass: 20 g — Example 2 - Dicyclohexylcarbodiimide (DCC) loading: 2.4 g — Example 2 - Milling temperature: 40° C. — Example 2 - Coupling/condensing agent amount: 1 to 12 parts by weight per hundred parts by weight of polymer — Claim 5 - Selective hydrogenation level: at least 80% of the carbon-carbon double bonds — Claim 6 - Catalyst loading: 0.01 to 1.0% by weight based on polymer — General synthesis - Preferred catalyst loading: 0.03% to 0.5% by weight based on polymer — General synthesis - More preferred catalyst loading: 0.06% to 0.12% by weight based on polymer — General synthesis - Most preferred catalyst loading: 0.08% by weight based on polymer — General synthesis - Co-catalyst loading: 0.3 to 5% by weight based on polymer — General synthesis - Preferred co-catalyst loading: 0.5 to 4% by weight based on polymer — General synthesis - Weight ratio of rhodium-containing catalyst to co-catalyst: 1:3 to 1:55 — General synthesis - More preferred weight ratio of rhodium-containing catalyst to co-catalyst: 1:5 to 1:45 — General synthesis - Co-catalyst weight based on 100 parts of rubber: 0.1 to 33 — General synthesis - Preferred co-catalyst weight based on 100 parts of rubber: 0.5 to 20 — General synthesis - More preferred co-catalyst weight based on 100 parts of rubber: 1 to 5 — General synthesis - Most preferred co-catalyst weight based on 100 parts of rubber: greater than 2 to less than 5 — General synthesis - Conjugated diene content: 40 to 85% by weight — General synthesis - Nitrile content: 14.9 to 50% by weight — General synthesis - Unsaturated acid content: 0.1 to 10% by weight — General synthesis - Preferred unsaturated acid content: 0.5 to 7% by weight — General synthesis - Another copolymerizable monomer content: not exceeding 10% — General synthesis - Average molecular weight: in excess of 60,000 g/mol — General synthesis - More preferred average molecular weight: in excess of 100,000 g/mol — General synthesis

Relevant Experimental Evidence - Example 1: HXNBR was milled with 0.9 g DCC at 40°C, then pressed at 150°C and 40,000 psi for 2 minutes; ATR showed loss of the isocyanate-like band and appearance of carbonyl bands, indicating derivatization/crosslinking. - Example 2: HXNBR was

milled at 40°C with 2.4 g DCC; ATR showed DCC residue and product carbonyl bands, demonstrating reaction under higher DCC loading.

Technical Relevance This patent is directly relevant because it addresses the chemistry of carboxylated nitrile rubber after carboxyl incorporation, especially HXNBR. It provides a practical route for crosslinking and derivatizing XNBR-derived materials without destroying the carboxyl functionality, which is important for property development and processing stability.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No qualifying primary patents were identified under the configured target material relevance criteria. Cross-patent quantitative trends were therefore not generated.

4. CONCLUSION

The evidence shows that XNBR-related technology in the manifest is focused on post-functionalization rather than direct carboxylated NBR synthesis. US20030109642A1 demonstrates that carboxylated nitrile rubbers, including hydrogenated variants, can be simultaneously crosslinked and derivatized using coupling/condensing agents such as dicyclohexylcarbodiimide. The most actionable disclosed conditions are milling at 40°C, DCC addition in the range of 0.9 to 2.4 g for 20 to 24 g polymer in the examples, and hot pressing at 150°C and 40,000 psi for 2 minutes. For synthesis and processing of XNBR-derived materials, the evidence supports selective hydrogenation of the diene double bonds while preserving nitrile and carboxyl groups, followed by carbodiimide-based derivatization/crosslinking to tune final properties.

5. REFERENCES

- US20030109642A1 | Process for crosslinking carboxylated nitrile rubber, hydrogenating ... | Not disclosed | US | 2003 | <https://patents.google.com/patent/US20030109642A1/en>